Nice RDF series for rolling shutters selection guide

Follow the arrow and choose the right product

For rolling shutters



Max. shutter weight (kg):

Gearmotor	Steel pipe EN 10220 (mm)										
RDF series	**101,6 x 3,6**	**108,0 x 3,6**	**133,0 x 4**	**159,0 x 4,5**	**177,8 x 5**	**193,7 x 5,4**	**219,1 x 5,9**	**244,5 x 6,3**	**273,0 x 6,3**	**298,5 x 7,1**	**323,9 x 7,1**
RDF-140-20	125	122	115	106	99	94	85	77	69	64	59
RDF 180-15	166	163	153	141	133	126	113	102	92	85	79
RDF-220-15	208	204	192	177	166	157	142	128	115	106	98
RDF-290-15	260	255	240	221	207	197	177	160	144	133	123
RDF-380-15	375	368	335	319	299	284	255	231	208	192	177
RDF-450-15	416	408	383	354	332	315	284	256	231	213	197
RDF-550-12	468	460	431	399	373	355	319	289	260	240	222
RDF-250-24	239	235	220	203	190	181	163	147	133	122	113
RDF-350-24	322	316	297	274	257	244	220	199	179	165	153
RDF-500-24	468	460	431	399	373	355	319	289	260	240	222
RDF-750-24	677	664	623	576	539	513	461	417	376	346	321
RDF-950-24	885	868	815	753	705	671	603	545	492	453	419
RDF-750-9.5	781	715	671	620	581	552	497	449	405	373	345
RDF-1250-24	1145	1124	1055	975	913	869	781	706	637	586	543
RDF-1400-24	1354	1324	1247	1152	1079	1026	923	835	753	693	642
RDF-850-10	-	-	-	664	622	592	532	481	434	400	370
RDF-1100-10	-	-	-	842	788	750	675	610	550	506	469
RDF-1400-7	-	-	-	1063	1038	987	888	770	724	666	617

The table applies only to section bars with a single wall (thickness 20 mm). The dimensions must be taken into account for the winding if section bars with double wall, with greater thickness or higher are used.

In this case higher torque applies after 1-2 turns.
A 20% loss of friction is considered for the normal rolling shutters in the table.

It is generally advisable to check the door proportions when there are coverings, section bars with double wall, doors higher than normal or in the case of particular entry situations that could require other force ratios or cause additional losses of friction (e.g. doors that are wider than the height), for example.

RDF



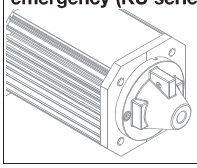
Indication of station choice to combine with the RDF gearmotors

	RDF series gearmotors				
Control unit	RDF-140-20 RDF-220-15 RDF-290-15	RDF-380-15 RDF-450-15 RDF-550-12	RDF-750-10 RDF-1000-10 RDF-1400-7	RDF-250-24 fino a RDF-750-24	RDF-950-24 RDF-1250-24 RDF-1400-24
UST2*	•	•			
UST1-5,5 kW					•
UST1-2,2 kW	•	•	•	•	
UST1K-5,5 kW					•
UST1K-2,2 kW	•	•	•	•	
UST1K-1,1 kW	•				

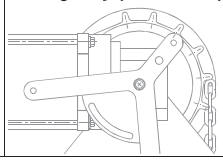
Nice RDF series gearmotors for rolling shutters



Handle-operated emergency (KU series)



Chain-operated emergency (KE2 series)



Electromechanical gearmotor for rolling shutters up to 750 kg, with integrated brake and mechanical or electronic limit switches, chain-operated emergency (KE2 series) or handle-operated emergency (KU series) with integrated parachute device.

Simple installation:
the gearmotor is to be installed directly on the shaft of the rolling shutter (Ø 30 mm, Ø 40 mm, Ø 55 mm). Connection cables equipped with connectors simplify electrical installation.

Complete Nice gearmotor-cable-station kits are directly wired in production and allow installation time to be further reduced.

Compact and efficient:
torque up to 550 Nm.
speed up to 12 rpm.

Unbeatable safety:
due to the antifall safety system and the chance to connect safety devices such as optical or resistive sensitive edges (8.2 kΩ). Gearmotors compliant with the most recent European standards and directives in terms of safety and technology.

Practical:
Release available in 2 versions
- handle-operated emergency
- chain-operated emergency, the ideal solution for every type of application.

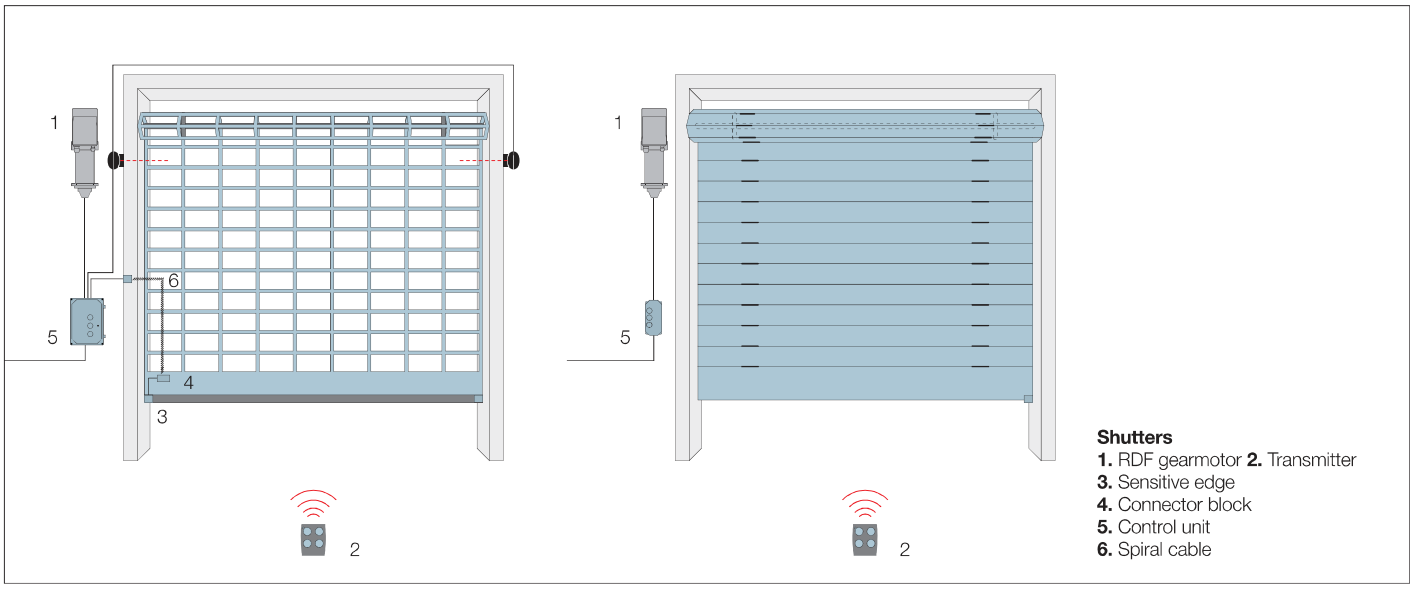
Available on request: special accessories for fastening, shafts and discs to be welded.

Complete preassembled kits

Code	Description
NDCK0375	Gearmotor with electronic limit switch and handle release, UST1 station with K2 and K4 modules and EEC plug, 5 m cable
NDCK0206	Gearmotor gearmotor with mechanical limit switch and handle release, UST1 station with K1 module and EEC plug, 5 m cable

Gearmotors

Code	Description
NDCM0079	RDF 220-15 gearmotor with mechanical limit switch and handle release, preassembled 7 m cable
NDCM0080	RDF 290-15 gearmotor with mechanical limit switch and handle release, preassembled 7 m cable
NDCM0110	RDF 380-15 gearmotor with mechanical limit switch and 5 m chain release, preassembled 7 m cable
NDCM0131	RDF 450-15 gearmotor with mechanical limit switch and handle release, preassembled 7 m cable
NDCM0149	RDF 550-12 gearmotor with mechanical limit switch and handle release, preassembled 7 m cable
NCDM0108	RDF 750-9.5 gearmotor with mechanical limit switch and 5 m chain release, preassembled 7 m cable
NDCM0033	RDF 1100-10 gearmotor with electronic limit switch and 5 m chain release, preassembled 9 m cable



Shutters
1. RDF gearmotor **2.** Transmitter
3. Sensitive edge
4. Connector block
5. Control unit
6. Spiral cable

Technical data

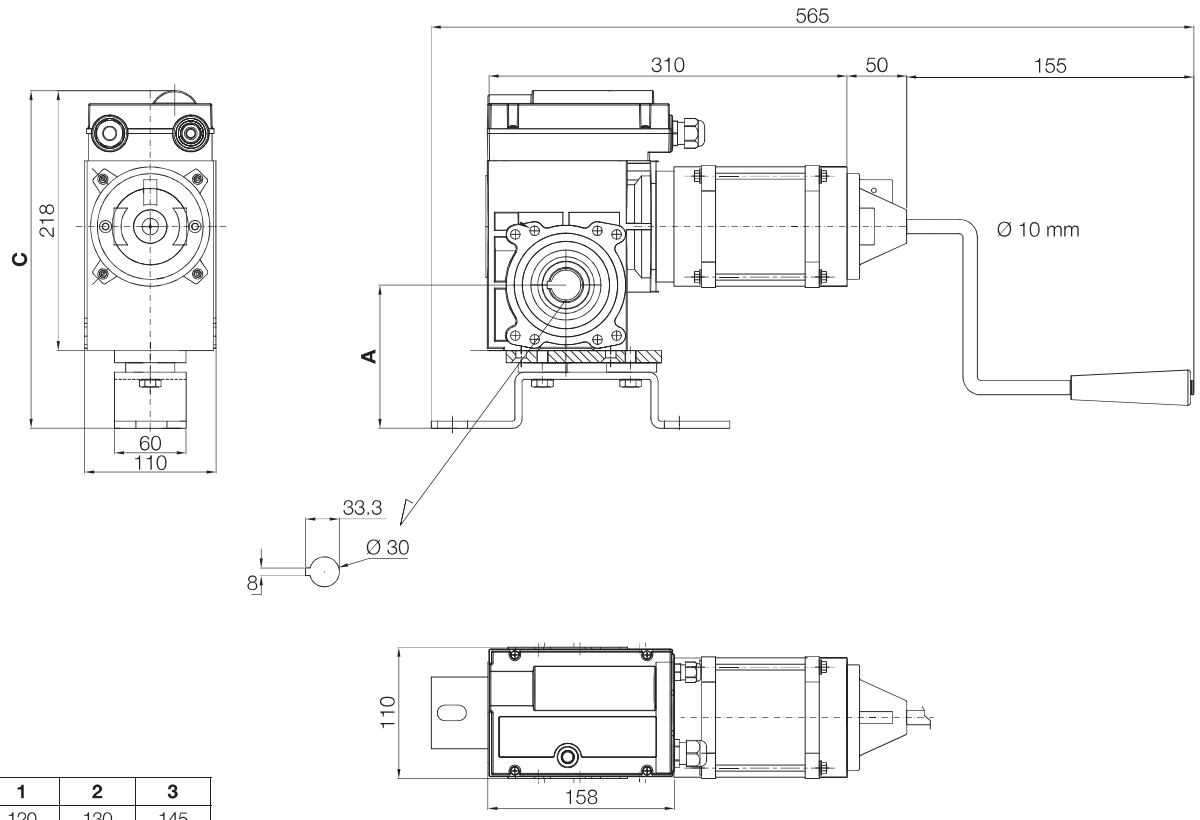
Model **RDF-**	**140-20**	**220-15**	**290-15**	**380-15**	**450-15**	**550-12**	**750-9,5**
Shaft Ø (mm)	30			40			
Max. torque (Nm)	140	220	290	380	450	550	750
Rated torque (Nm)	120	200	250	360	400	450	700
Output rpm	20	15	15	15	15	12	9,5
Motor power (kW)	0,55	1,1	1,1	1,5	1,5	1,5	1,5
Limit switch rpm	15	10-20	10-20	10-20	10-20	10	40
Working voltage (V/Hz)	3x230 V / 3x400 V 50 Hz						
Duty Cycle	S 3 - 60%						
Connection cable (no. x mm^2)	5 x 1,5						
Operating temperature (°C)	-5 ÷ +40						
Noise dB(A)	<70						
Protection rating (IP)	IP 54						
Weight (kg)	12,4	22	24	28,9	27,7	27,7	35

Model **RDF-**	**850-10**	**1100-10**	**1400-7**	**250-24**	**350-24**	**500-24**	**750-24**	**950-24**	**1250-24**	**1400-24**
Shaft Ø (mm)	55			40			55			
Max. torque (Nm)	850	1100	1400	250	350	500	750	950	1250	1400
Rated torque (Nm)	700	950	1250	230	310	450	650	850	1100	1250
Output rpm	10	10	7	24	24	24	24	24	24	24
Motor power (kW)	2,2	2,2	2,2	1,1	1,5	2,2	2,2	3	4	5,5
Limit switch rpm	10-40	10-40	10-40	10-40	10-40	10-40	10-40	10-40	10-40	10-40
Working voltage (V/Hz)	3x230 V / 3x400 V 50 Hz									
Duty Cycle	S 3 - 60%									
Connection cable (no. x mm^2)	5 x 1,5									
Operating temperature (°C)	-5 ÷ +40									
Noise dB(A)	<70									
Protection rating (IP)	IP 54									
Weight (kg)	51,7	51,9	58	24,1	29,2	32,7	51,9	61	63	65

Gearmotors with integrated parachute device, independent of speed and position.

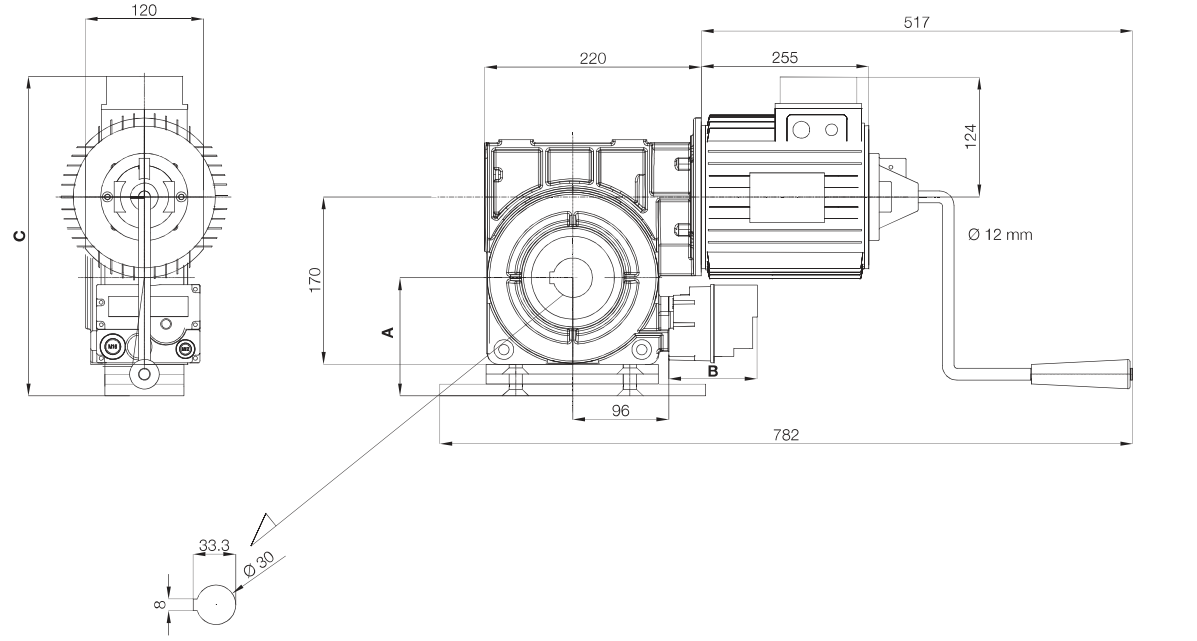
Nice RDF series technical drawings for rolling shutters

RDF-140-20-KU



Dimension	1	2	3
A	120	130	145
C	284	294	309

RDF-220-15-KU

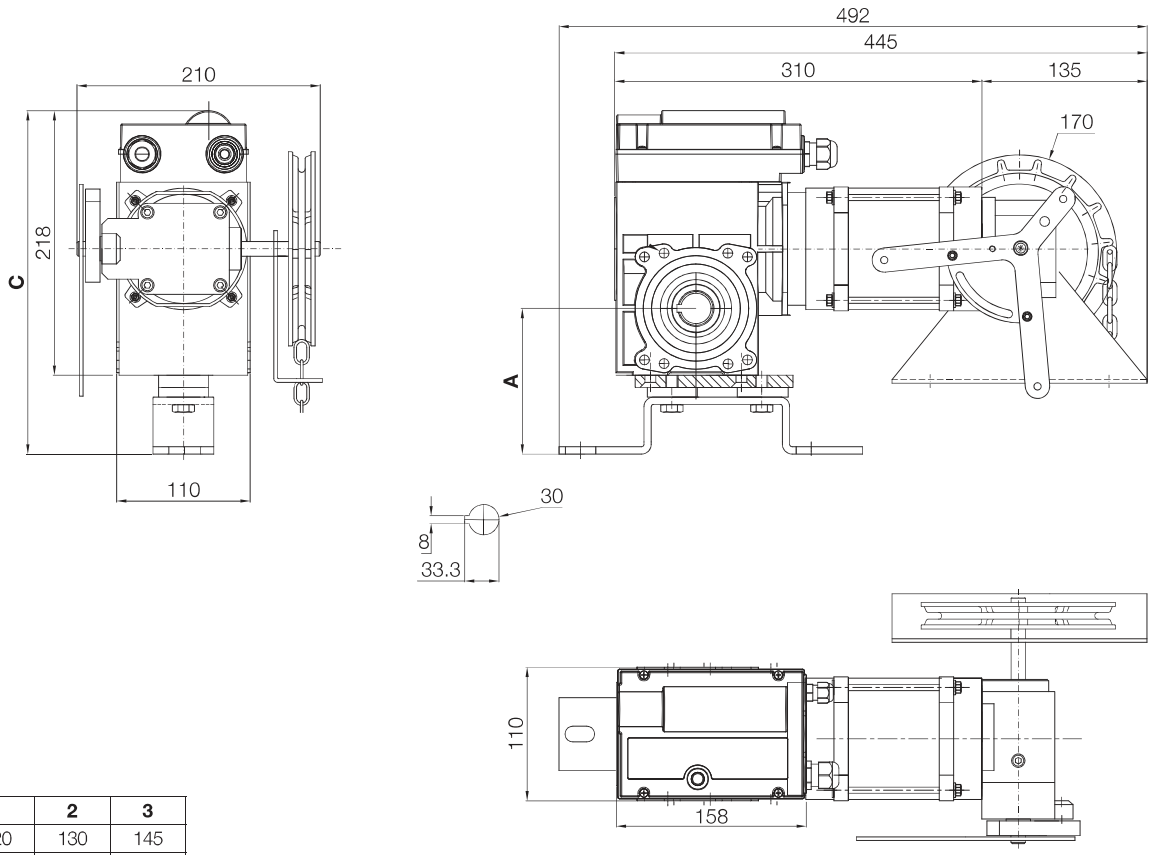


Dimension	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	326	336	351

B = 90 electronic limit switches
B = 150 mechanical limit switches

Key
Dimension A: shaft centre - fixing bracket lower edge distance | **Dimension B:** limit switch housing length | **Dimension C:** gearmotor construction height | **1, 2, 3:** possibilities available

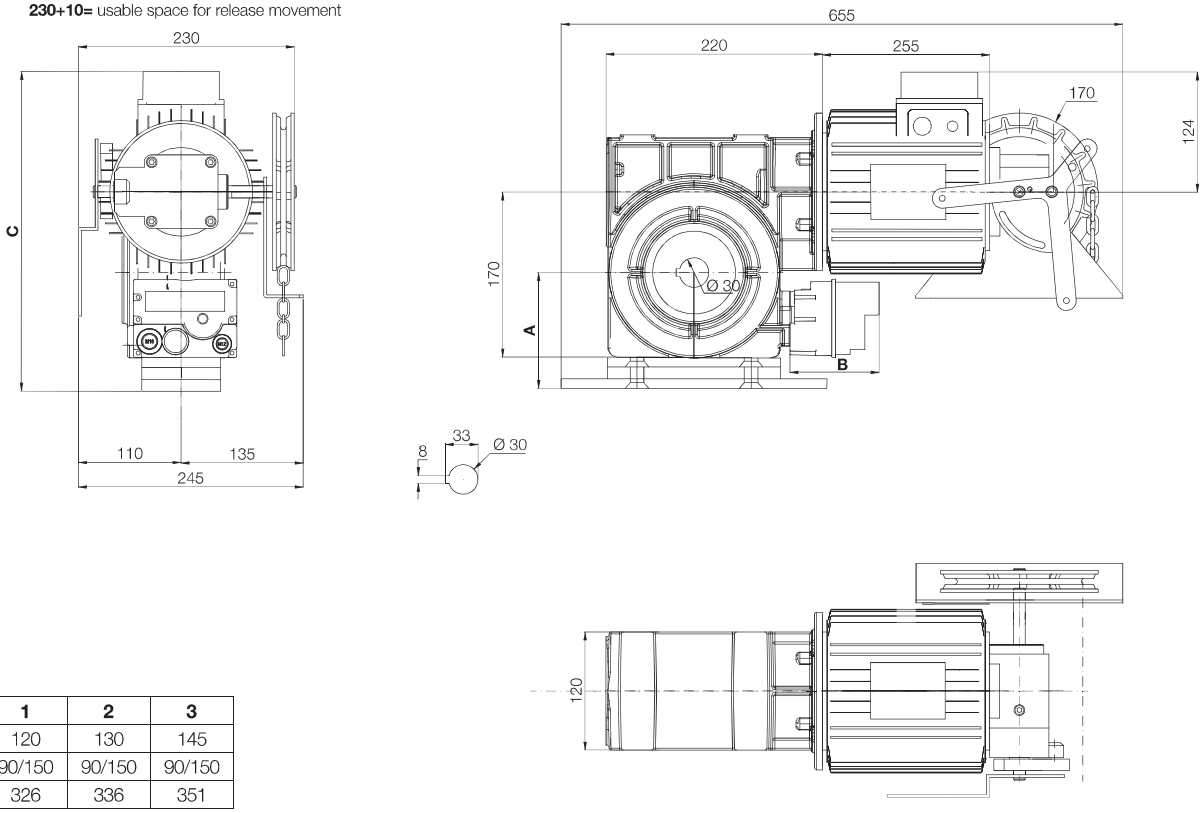
RDF-140-20-KE2



Dimension	1	2	3
A	120	130	145
C	284	294	309

RDF-220-15-KE2

230+10= usable space for release movement

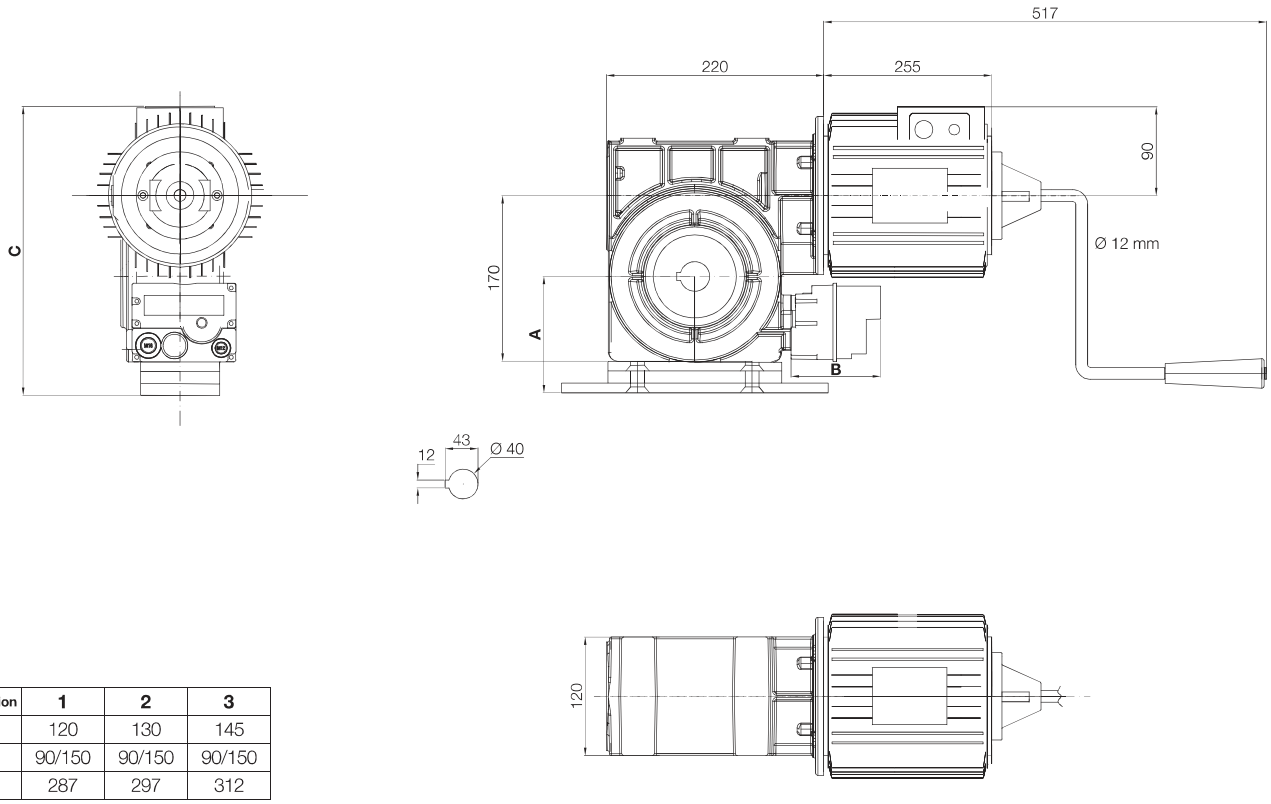


Dimension	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	326	336	351

B = 90 electronic limit switches
B = 150 mechanical limit switches

Nice RDF series technical drawings for rolling shutters

RDF-380/450-15-KU

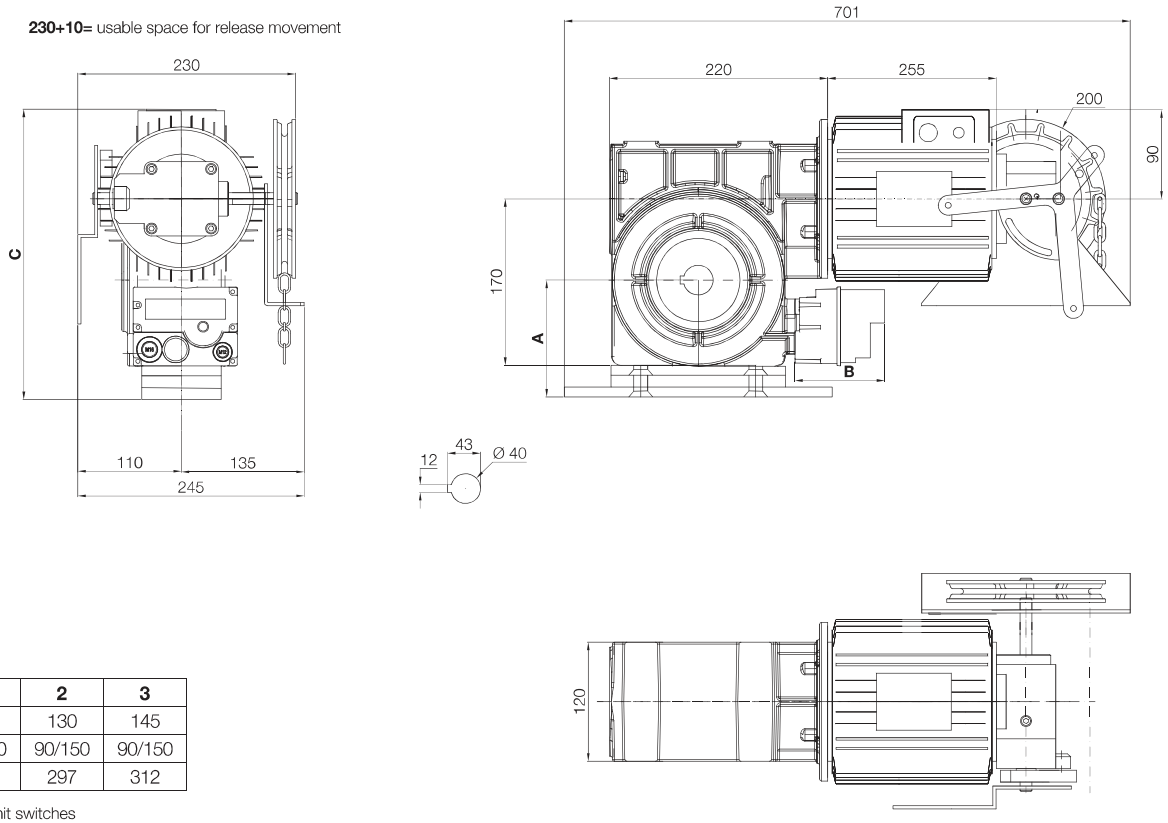


Dimension	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	287	297	312

B = 90 electronic limit switches
B = 150 mechanical limit switches

RDF-380/450-15-KE2

230+10= usable space for release movement

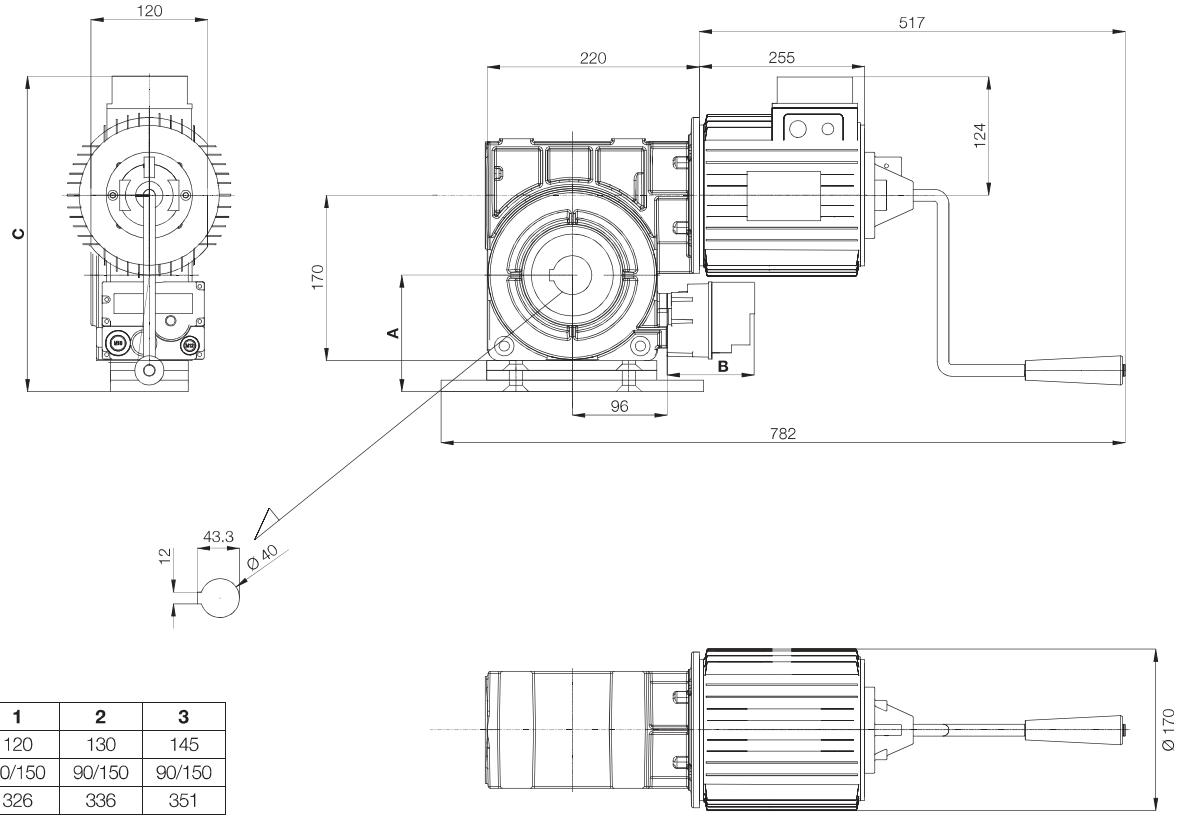


Dimension	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	287	297	312

B = 90 electronic limit switches
B = 150 mechanical limit switches

Key
Dimension A: shaft centre - fixing bracket lower edge distance | **Dimension B:** limit switch housing length | **Dimension C:** gearmotor construction height | **1, 2, 3:** possibilities available

RDF-550-12-KU

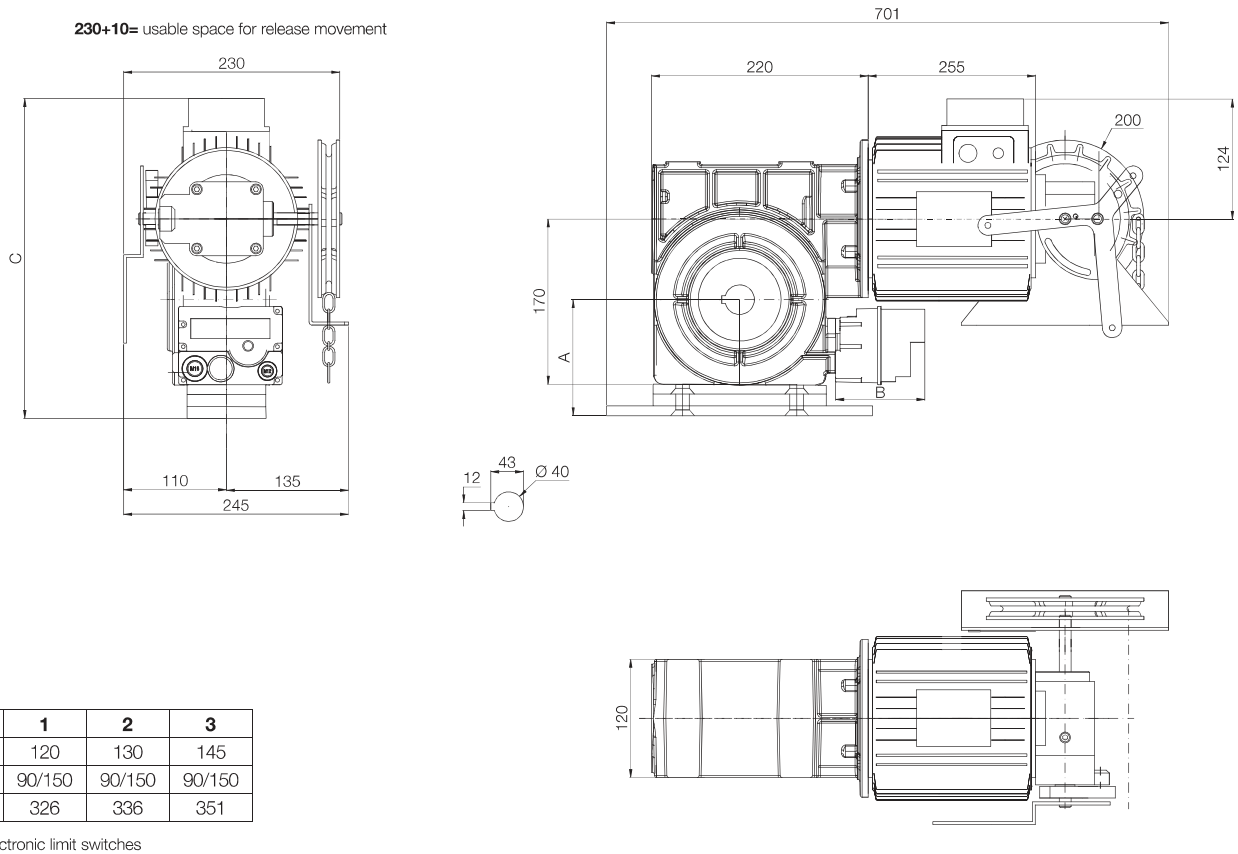


Dimension	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	326	336	351

B = 90 electronic limit switches
B = 150 mechanical limit switches

RDF-550-12-KE2

230+10= usable space for release movement

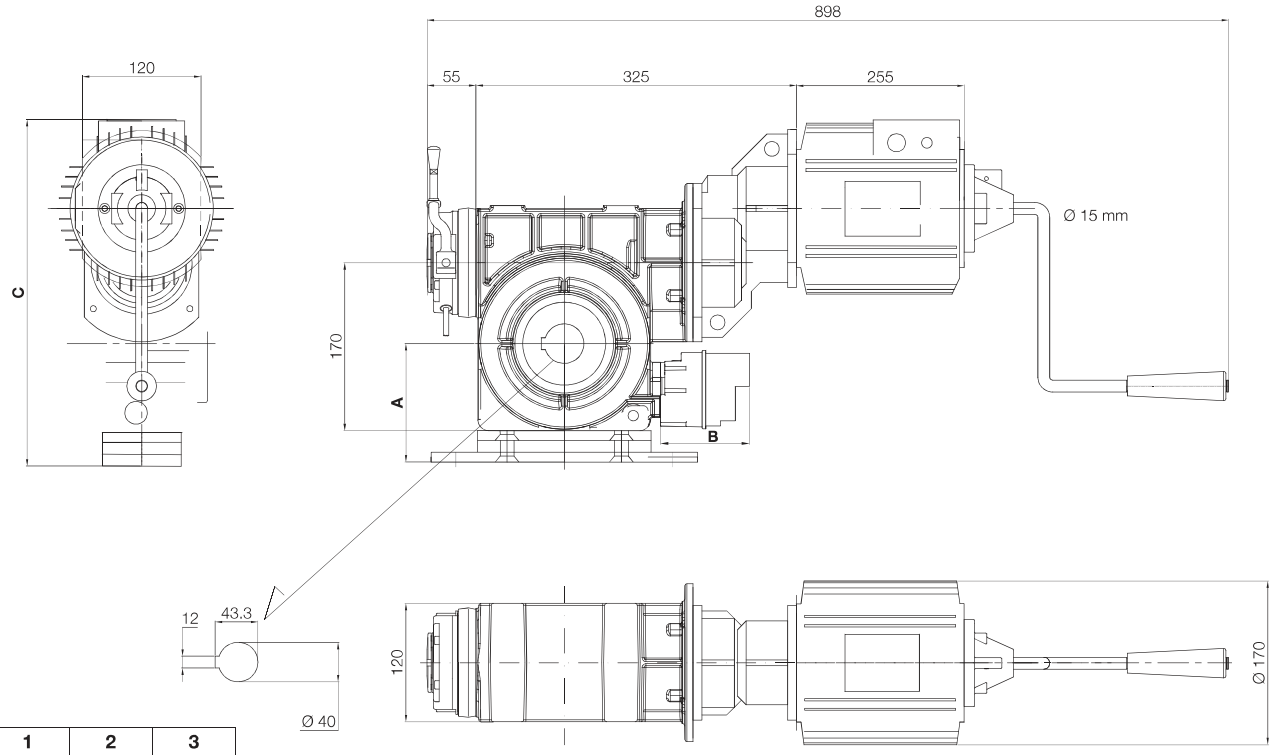


Dimension	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	326	336	351

B = 90 electronic limit switches
B = 150 mechanical limit switches

Nice RDF series technical drawings for rolling shutters

RDF-750-9.5-KU

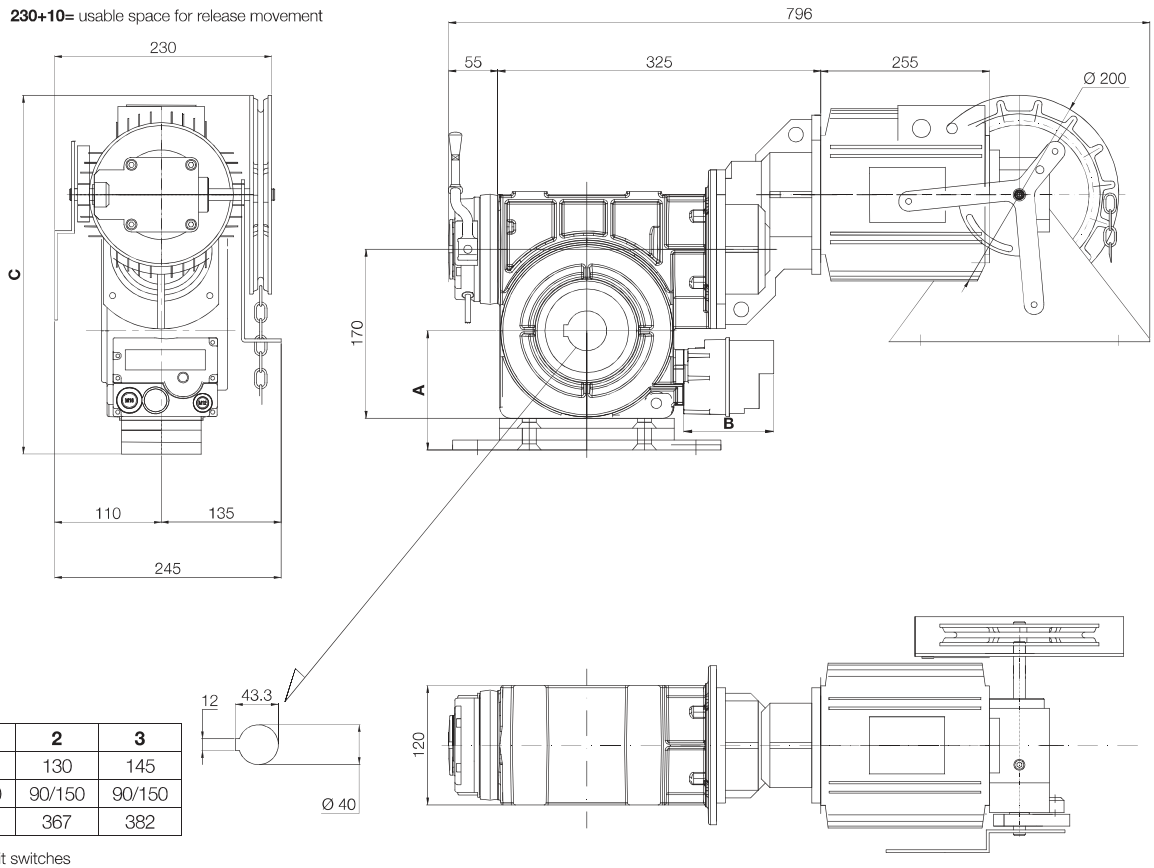


Dimension	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	347	357	372

B = 90 electronic limit switches
B = 150 mechanical limit switches

RDF-750-9.5-KE2

230+10= usable space for release movement

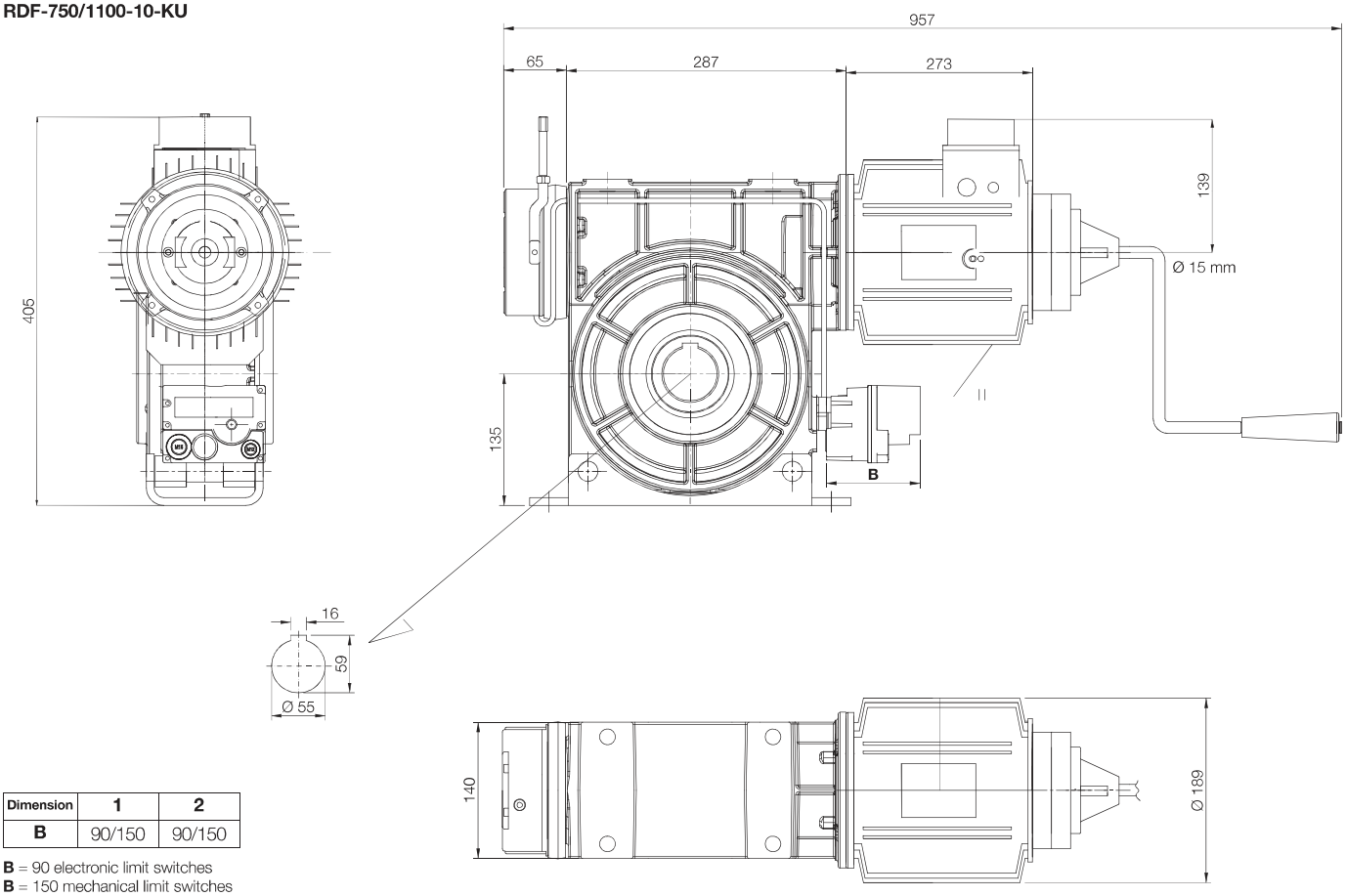


Dimension	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	357	367	382

B = 90 electronic limit switches
B = 150 mechanical limit switches

Key
Dimension A: shaft centre - fixing bracket lower edge distance | **Dimension B:** limit switch housing length | **Dimension C:** gearmotor construction height | **1, 2, 3:** possibilities available

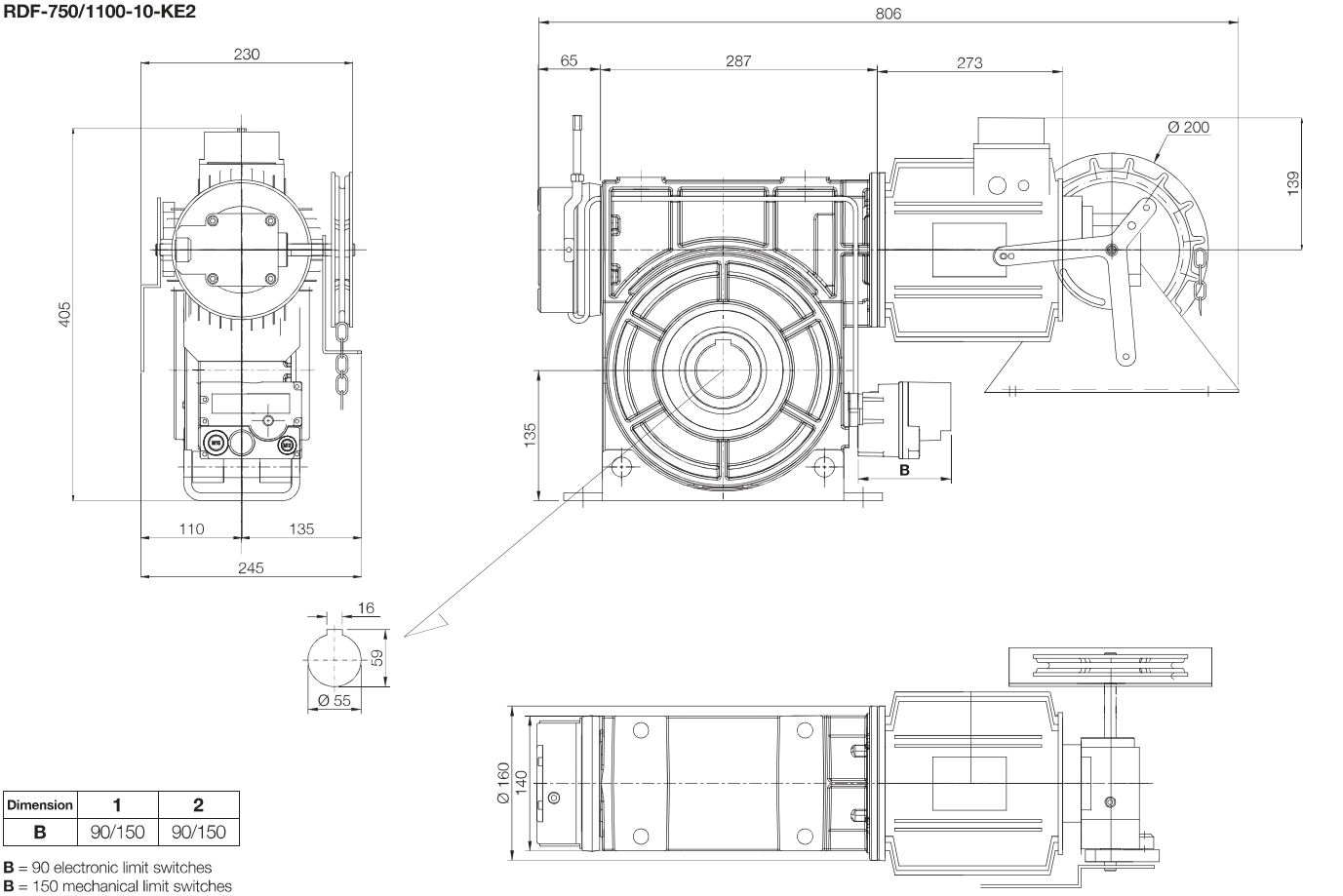
RDF-750/1100-10-KU



Dimension	1	2
B	90/150	90/150

B = 90 electronic limit switches
B = 150 mechanical limit switches

RDF-750/1100-10-KE2

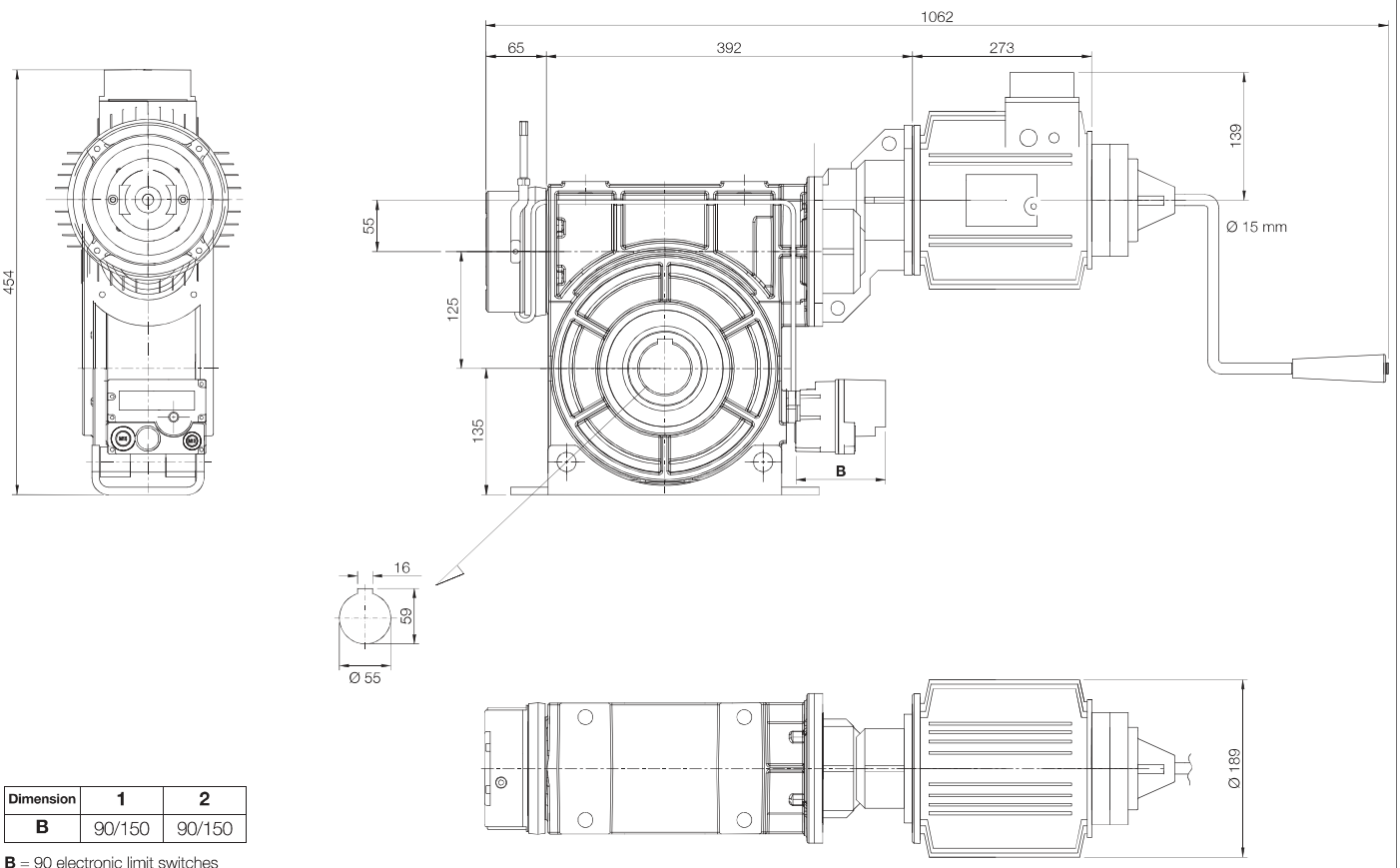


Dimension	1	2
B	90/150	90/150

B = 90 electronic limit switches
B = 150 mechanical limit switches

Nice RDF series technical drawings for rolling shutters

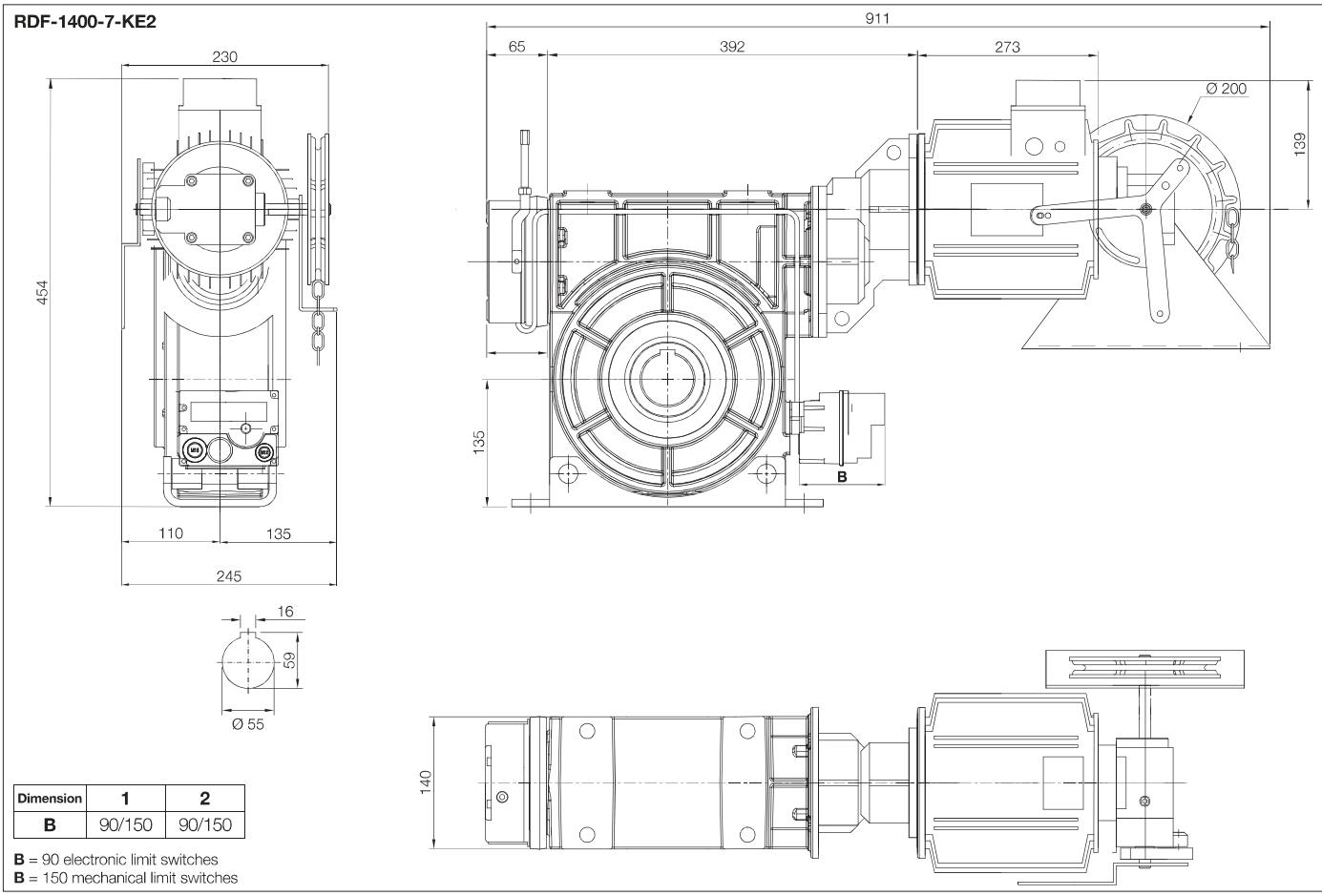
RDF-1400-7-KU



Dimension	1	2
B	90/150	90/150

B = 90 electronic limit switches
B = 150 mechanical limit switches

RDF-1400-7-KE2



Dimension	1	2
B	90/150	90/150

B = 90 electronic limit switches
B = 150 mechanical limit switches